Inequalities — problem 7.84

A. Engel — Problem-Solving Strategies · 1998

Let $a_1, \ldots, a_n$ be positive and $a_{n+1} = a_1$. Prove that

$$2\sum_{k=1}^{n} \frac{a_k^2}{a_k + a_{k+1}} \geq \sum_{k=1}^{n} a_k.$$